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Abbreviations for Amino Acids

A	Ala	Alanine	N	Asn	Asparagine
B	Asx	Asparagine or aspartate	P	Pro	Proline
			Q	Gln	Glutamine
C	Cys	Cysteine	R	Arg	Arginine
D	Asp	Aspartate	S	Ser	Serine
E	Glu	Glutamate	T	Thr	Threonine
F	Phe	Phenylalanine	V	Val	Valine
G	Gly	Glycine	W	Trp	Tryptophan
H	His	Histidine	X	—	Unknown or nonstandard amino acid
I	Ile	Isoleucine			
K	Lys	Lysine	Y	Tyr	Tyrosine
L	Leu	Leucine	Z	Glx	Glutamine or glutamate
M	Met	Methionine			

Asx and Glx are used in describing the results of amino acid analytical procedures in which Asp and Glu are not readily distinguished from their amide counterparts, Asn and Gln.

The Standard Genetic Code

UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys
UUC	Phe	UCC	Ser	UAC	Tyr	UGC	Cys
UUA	Leu	UCA	Ser	UAA	Stop	UGA	Stop
UUG	Leu	UCG	Ser	UAG	Stop	UGG	Trp
CUU	Leu	CCU	Pro	CAU	His	CGU	Arg
CUC	Leu	CCC	Pro	CAC	His	CGC	Arg
CUA	Leu	CCA	Pro	CAA	Gln	CGA	Arg
CUG	Leu	CCG	Pro	CAG	Gln	CGG	Arg
AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser
AUC	Ile	ACC	Thr	AAC	Asn	AGC	Ser
AUA	Ile	ACA	Thr	AAA	Lys	AGA	Arg
AUG	Met*	ACG	Thr	AAG	Lys	AGG	Arg
GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly
GUC	Val	GCC	Ala	GAC	Asp	GGC	Gly
GUA	Val	GCA	Ala	GAA	Glu	GGA	Gly
GUG	Val	GCG	Ala	GAG	Glu	GGG	Gly

*AUG also serves as the initiation codon in protein synthesis.

1 H 1.008												2 He 4.0							
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.011	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18		
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95		
19 K 39.10	20 Ca 40.08			21 Sc 44.96	22 Ti 47.90	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.71	29 Cu 63.55	30 Zn 65.37	31 Ga 69.72	32 Ge 72.59	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.30
37 Rb 85.47	38 Sr 87.62			39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Te 98.91	44 Ru 101.07	45 Rh 102.91	46 Pd 106.4	47 Ag 107.87	48 Cd 112.40	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 126.70	53 I 126.90	54 Xe 131.30
55 Cs 132.91	56 Ba 137.34	57–70 *	71 Lu 174.97	72 Hf 178.49	73 Ta 180.95	74 W 183.85	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.09	79 Au 196.97	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)	
87 Fr (223)	88 Ra 226.03	89–102 **	103 Lr 262.11	104 Rf 261.11	105 Db 262.11	106 Sg 263.12	107 Bh 264.12	108 Hs 265.13	109 Mt 268	110 Ds 281	111 Rg 281	112 Cn 285	113 Nh 286	114 Fl 289	115 Mc 289	116 Lv 293	117 Ts 293	118 Og 294	

*Lanthanides

57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm 144.91	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04
89 Ac 227.03	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.05	94 Pu 244.06	95 Am 243.06	96 Cm 247.07	97 Bk 247.07	98 Cf 251.08	99 Es 252.08	100 Fm 257.10	101 Md 258.10	102 No 259.10

**Actinides

Bioinformatics Databases

National Center for Biotechnology Information (NCBI)
 UniProt
 ExPASy Bioinformatics Resource Portal
 GenomeNet

www.ncbi.nlm.nih.gov
www.uniprot.org
www.expasy.org
www.genome.jp

Structure Data Banks and Databases

Protein Data Bank (PDB)
 EMDataBank Unified Resource for 3DEM
 National Center for Biomedical Glycomics
 LIPIDMAPS Lipidomics Gateway
 Nucleic Acid Database (NDB)
 Modomics database of RNA modification pathways

www.rcsb.org
www.emdataresource.org
www.glycomics.uga.edu
www.lipidmaps.org
www.ndbserver.rutgers.edu
www.genesilico.pl/modomics

Other Resources and Tools

Structural Classification of Proteins database (SCOP2)
 PROSITE Sequence logo
 ProtScale hydrophobicity and other profiles of amino acids
 Predictor of Natural Disordered Regions (PONDR)
 Enzyme nomenclature
 Ensembl genome databases
 PANTHER (Protein ANALysis THrough Evolutionary Relationships)
 Classification System
 Basic Local Alignment Search Tool (BLAST)
 Kyoto Encyclopedia of Genes and Genomes (KEGG)
 KEGG pathway maps
 Biochemical nomenclature
 Online Mendelian Inheritance in Man

<http://scop2.mrc-lmb.cam.ac.uk>
http://prosite.expasy.org/sequence_logo.html
<http://web.expasy.org/protscale>
www.pondr.com
www.qmul.ac.uk/sbcs/iubmb/enzyme
www.ensembl.org
www.pantherdb.org
<https://blast.ncbi.nlm.nih.gov/Blast.cgi>
www.genome.jp/kegg
www.genome.ad.jp/kegg/pathway/map/map01100.html
www.qmul.ac.uk/sbcs/iubmb
www.omim.org

Unit Abbreviations

A	ampere	L	liter
Å	angstrom	M	molar (concentration)
atm	atmosphere	m	meter
°C	degree Celsius	min	minute
cal	calorie	mm Hg	millimeters of mercury (pressure)
cpm	counts per minute	mol	mole
Da	dalton	N	normal (concentration)
dpm	disintegrations per minute	Pa	pascal
F	faraday	r	revolution
g	gram	S	Svedberg unit
h	hour	s	second
J	joule	V	volt
K	kelvin	yr	year

Prefixes

10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f

Some Conversion Factors

Length	$1 \text{ cm} = 10 \text{ mm} = 10^4 \mu\text{m} = 10^7 \text{ nm}$ $= 10^8 \text{ Å} = 0.394 \text{ in}$ $1 \text{ in} = 2.54 \text{ cm}$ $1 \text{ yard} = 0.9144 \text{ meters}$ $1 \text{ mile} = 1.609 \text{ kilometers}$
Mass	$1 \text{ g} = 10^{-3} \text{ kg} = 10^3 \text{ mg} = 10^6 \mu\text{g}$ $= 3.53 \times 10^{-2} \text{ oz}$ $1 \text{ oz} = 28.3 \text{ g}$
Temperature	$^{\circ}\text{C} = 5/9(^{\circ}\text{F} - 32)$ $\text{K} = ^{\circ}\text{C} + 273$
Energy	$1 \text{ J} = 10^7 \text{ erg} = 0.239 \text{ cal}$ $1 \text{ cal} = 4.184 \text{ J}$
Pressure	$1 \text{ torr} = 1 \text{ mm Hg} = 1.32 \times 10^{-3} \text{ atm}$ $= 1.333 \times 10^2 \text{ Pa}$ $1 \text{ atm} = 758 \text{ torr} = 1.01 \times 10^5 \text{ Pa}$

Some Physical Constants, with Symbols and Values

Atomic mass unit (dalton)	amu	$1.661 \times 10^{-24} \text{ g}$
Avogadro's number	N	$6.022 \times 10^{23} / \text{mol}$
Boltzmann constant	k	$1.381 \times 10^{-23} \text{ J/K};$ $3.298 \times 10^{-24} \text{ cal/K}$
Faraday constant	F	$96,480 \text{ J/V} \cdot \text{mol}$
Gas constant	R	$1.987 \text{ cal/mol} \cdot \text{K};$ $8.315 \text{ J/mol} \cdot \text{K}$
Planck's constant	h	$1.584 \times 10^{-34} \text{ cal} \cdot \text{s};$ $6.626 \times 10^{-34} \text{ J} \cdot \text{s}$
Speed of light (in vacuum)	c	$2.998 \times 10^{10} \text{ cm/s}$





C

C

C

C

C

C

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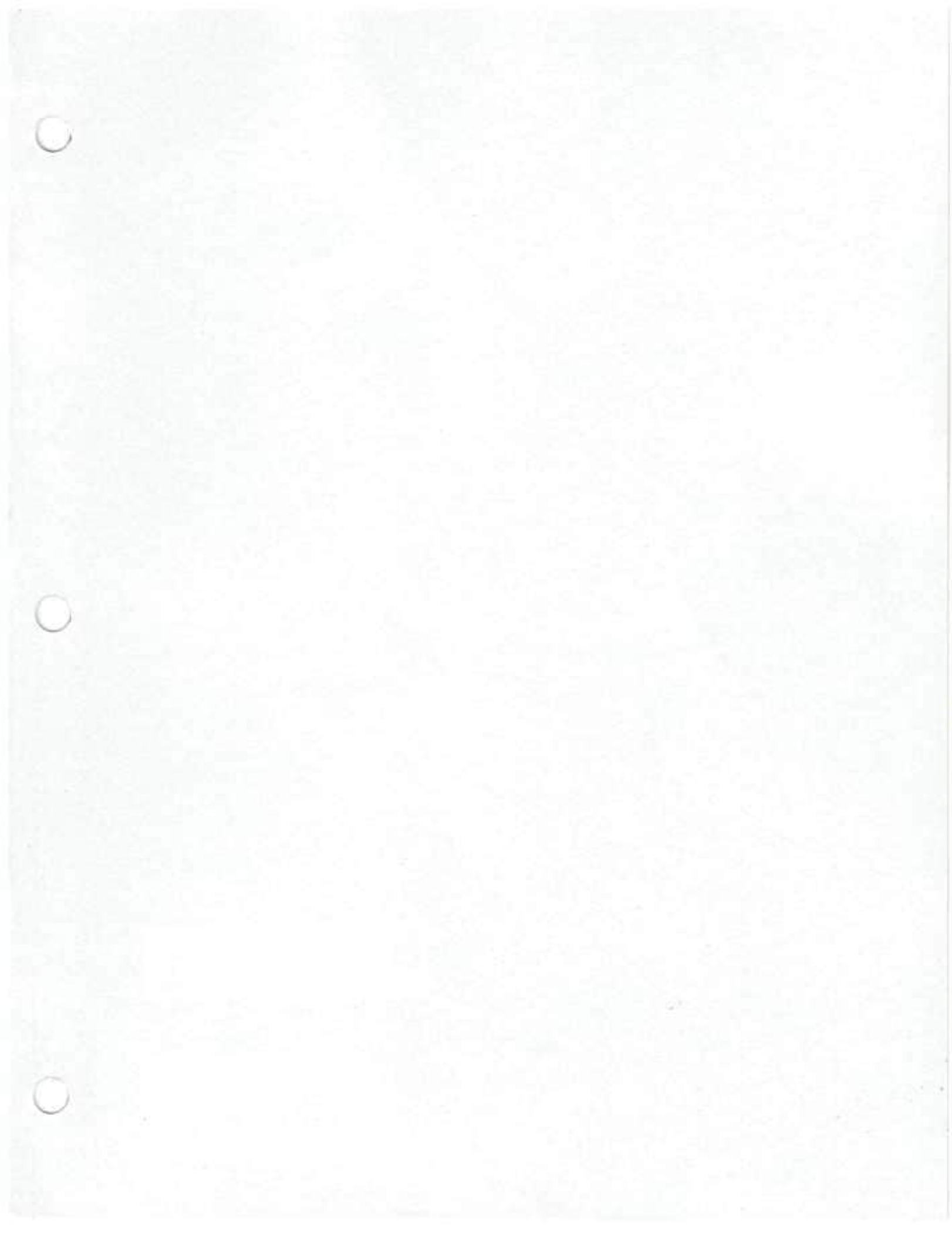
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ON THE COVER:

In this artistic rendering, a protocell composed of a bilayer of fatty acids surrounds an aqueous compartment containing an RNA ribozyme and nucleic acids. Protocells are used by biochemists to explore reactions that could mimic those that led to the origin of life on Earth. Billions of years ago, hydrophobic molecules may have formed cell-like assemblies like protocells in which molecules capable of replicating themselves could be concentrated and protected from the external environment. These likely provided the origins of the membranes, nucleic acids, and catalysts found in today's cells.

**BUDGET
BOOKS**

This book does not have a traditional binding; its pages are loose and 3-hole punched to provide flexibility and a low price to students.

Cover image: Janet Iwasa, University of Utah



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ISBN-13: 978-1-319-38147-9
ISBN-10: 1-319-38147-2



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